

Content requirement

Diagram on the left indicates where learners may apply mathematical skills or science knowledge.

The 'explore', 'create' and 'evaluate' column uses different size dots, not only to indicate the interrelationship, but also their significance to any topic strand.

Before to undertake their Iterative Design Challenge, learners should:

Identify requirements

	Explore	Create	Evaluate
a. Understand that all design and technological practice takes place within contexts that inform outcomes. Learners should be able to identify and prioritise problems and opportunities that are relevant to their chosen context. These issues should be reflected on throughout their project.	●	●	●
b. Be able to write a design brief in response to a contextual challenge that considers the stakeholders that could have an interest in the potential outcome.	●	●	●
c. Investigate the needs, wants and interests of primary users and other stakeholders to identify and understand the requirements for designing, through collecting, analysing and presenting their findings from primary and secondary data.	●	●	●

Learning from existing products and practice

a. Investigate and analyse relevant existing products, understanding how they are used within their physical, organisational, social and/or cultural environments, using methods such as disassembly and systems thinking in order to make informed and reasoned decisions.	●	●	●
b. Investigate and analyse the wider work of professionals and			

Applications of wider issues		Explore	Create	Evaluate
a.	Investigate factors, such as environmental, social and economic challenges, in order to identify opportunities and constraints that might influence the process of iterative designing and making, such as taking into consideration the ecological and social footprint of materials.	●	●	●
Design thinking and communication				
a.	Demonstrate an ability to identify and formulate appropriate requirement lists and specifications, reflecting on their own investigations and considering stakeholder needs, including: - requirements that cover stakeholder needs and wants - requirements that cover technical needs - technical specifications that outline the specific requirements needed to support the design solution being made into a final prototype.	●	●	●
b.	Be able to use different design strategies and approaches such as collaboration, user-centred design and systems thinking when generating and developing innovative design ideas that avoid design fixation.	●	●	●
c.	Be able to design and develop at least one prototype that responds to needs and/or wants and is fit for purpose, demonstrating functionality, aesthetics and marketability.	●	●	●
d.	Apply techniques in order to communicate and record design ideas suitable to the stage of development in order to justify their own thinking and present their thinking and intentions to a third party, such as: - informal 2D and 3D sketching and modelling to communicate initial ideas - system and schematic diagrams, annotated sketches, exploded diagrams, models and written notes, to communicate development iterations - audio and visual recordings to share thinking, explorations and the functionality of ideas - formal 2D and 3D working drawings to outline specification requirements; 3D illustrations, mathematical modelling and computer-based tools to present final design solutions; schedules and flow charts to deliver planning - presentations and real-time evidence to communicate throughout the project.	●	●	●

Material considerations		Explore	Create	Evaluate
a.	Develop and apply in-depth knowledge by selecting and working with appropriate materials and components when developing their ideas, early models and producing their final prototype(s).	●	●	●
Technical understanding				
a.	Apply technical principles appropriately to ensure functional requirements are achieved when developing a design solution.	●	●	●
b.	Be able to use surface treatments and finishes for functional and aesthetic purposes.	●	●	●
Manufacturing processes and techniques				
a.	Be able to use specialist techniques and processes to shape, fabricate, construct and assemble at least one high quality prototype, including techniques such as wastage, addition, deforming and reforming, as appropriate to the materials and/or system components being used.	●	●	●
b.	Be able to use specialist tools and equipment, appropriate to the materials or system components used (including hand tools, machinery, digital design and manufacture), to create models and prototypes.	●	●	●
c.	Be able to use appropriate and accurate marking out methods including: measuring and use of reference points, lines and surfaces; use templates, jigs and/or patterns where appropriate; work within tolerances; understand efficient cutting and how to minimise waste.	●	●	●
Ability of design solutions				
a.	Be able to test, critically analyse and evaluate their design solutions against the identified stakeholder requirements, design opportunities and constraints in order to refine and improve future iterations.	●	●	●
b.	Be able to make informed and reasoned decisions throughout the iterative design process, responding to feedback as appropriate to identify the potential next steps of further development.	●	●	●
c.	Be able to respond to feedback given by others about their prototype(s) in order to identify the potential for further			